Sahaj Bhadja

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Summary

Software engineer with 2+ years of experience validating and shipping reliable software across **Python, C++/JavaScript, SQL, and Linux/Windows environments**, with strong focus on **functional, regression, and integration testing**, defect triage, and automation-assisted quality workflows.

Education

University of Southern California **Los Angeles, CA**
Master of Science in Computer Science *August 2024 – May 2026*
GPA: 3.75/4 Coursework: Analysis of Algorithms, Database Systems, Web Technologies

PDEU
B.Tech. in Computer Science and Engineering *Nov 2020 – May 2024*
GPA: 3.88/4 Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*
- Designed and validated backend workflows across **FastAPI**, **Azure SQL**, and **PostgreSQL** for a multi-tenant platform serving 500+ users, improving response reliability and reducing dashboard latency from 1.2s to 350ms.
- Executed regression-oriented validation for 5K+ monthly background tasks with reconciliation checks, error handling, and release safeguards, sustaining 99%+ production reliability.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*
- Built and tested AI product workflows with **Flask** + **React**, secure file ingestion, and modular APIs, supporting 10K+ daily requests across multiple build and runtime scenarios.
- Implemented concurrent processing and CI/CD-backed release checks on **AWS RDS**, reducing turnaround time by 60% while improving stability under peak-load test conditions.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Developed and executed 20+ unit/integration suites with **Jenkins**, **PyTest**, and Mockito across Windows, Linux, and macOS, improving repeatability of test execution and defect detection.
- Partnered on API and ETL issue triage using logs, reproducible scenarios, and OOP-based fixes across **AWS Glue/S3** workflows, improving response times by 30% and pipeline efficiency by 68%.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Validated REST API and WebSocket behaviors for a microservice-based operations platform supporting 8+ business workflows and 2M+ synchronized records, ensuring reliable end-user flows across releases.
- Supported structured deployment and retest cycles using **Jenkins**, **Docker**, and Azure SQL-backed services, reducing deployment time by 50% and sustaining 95%+ successful build validation.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built Python-based simulation and evaluation pipelines for autonomous systems, processing 100K+ trajectory logs and increasing long-horizon planning performance by 18% through iterative test-and-debug cycles.
- Documented model behavior, diagnostics, and reproducible experiment results to improve debugging clarity and support release-readiness style reporting across research iterations.

Skills

Languages: Python, JavaScript/TypeScript, Java, SQL, C/C++
Testing & QA: Functional Testing, Regression Testing, Integration Testing, Exploratory Testing, Defect Lifecycle, Root-Cause Triage
Automation & Tooling: PyTest, CI Pipelines, Jenkins, Git/GitHub, Test Evidence and Repro Documentation, Jira/TestRail Concepts
Platforms: Linux, Windows, Command-Line Tooling, Log-Based Debugging, Serial/Device Log Analysis Concepts
Backend & APIs: FastAPI, Flask, Node.js, Express.js, REST APIs, JSON, WebSocket Validation
Databases: PostgreSQL, MySQL, MongoDB, SQL Server, Azure SQL, Relational Data Modeling
Engineering Practices: SDLC, Agile/Scrum, Risk Communication, Release Readiness Reporting, Cross-Functional Collaboration

Projects

CPU-Scheduling Simulator | *C++, React.js, JavaScript*
- Implemented and verified 6 scheduling algorithms with live visualizations across 1K+ simulated process combinations, reducing debugging time by 20% through repeatable test scenarios.

HMIS | *MERN, ABDM APIs, TypeScript, Twilio*
- Built and validated a healthcare workflow platform with consent-based API flows and structured monitoring, supporting 100+ daily appointment workflows while reducing record retrieval time by 2x.

Chat2SQL | *Java, TypeScript, SQL, Docker, Maven*
- Built an AI-assisted SQL workspace across a Java + TypeScript stack, supporting 10+ databases and structured query execution with reproducible validation across environments.